

Project: PlateWise AI

Next-Generation Multimodal Nutrition & Health Assistant

1. Project Vision

PlateWise is not just an app; it is a sophisticated **Predictive Health Ecosystem**. In a world of rising lifestyle-related health issues, generic advice is no longer enough. Our mission is to build an intelligent, vision-aware assistant that understands the unique relationship between a specific user and the food on their plate. By combining **Computer Vision (CV)** and **Natural Language Processing (NLP)**, students will build a system that can "see" a meal, decompose it into its nutritional parts, and provide real-time, goal-oriented advice through a seamless conversational interface. This project challenges students to build a prototype that moves beyond static datasets and build a living, integrated AI pipeline.

2. Project Description

Imagine a user standing in a cafeteria. They have a goal to manage their sodium intake. They take a photo of a laksa. Within seconds, the **PlateWise** engine:

1. **Segments** the bowl from the tray.
2. **Identifies** the noodles, prawns, and coconut-based broth.
3. **Estimates** the portion size and calculates the likely sodium and calorie count.
4. **Cross-references** this data with the user's stored health profile.
5. **Engages** in a dialogue: *"This looks delicious, but the broth is high in sodium. Perhaps enjoy the prawns and noodles, but skip half the soup to stay on track for today?"*

3. Technical Execution & Research Tasks

The project begins with the development of an **Intelligent Onboarding System**. The team must design a conversational interface using **Natural Language Processing (NLP)** to conduct a structured interview with the user. The goal is to establish a robust User Health Profile by capturing biometrics, allergies, and specific dietary objectives. This requires the implementation of dialogue management techniques to ensure the chatbot accurately parses and stores user intent, creating the foundation for all subsequent personalised advice.

Once the profile is established, the focus shifts to **Visual Meal Decomposition**. This task requires the application of **Computer Vision**, specifically the branch of **Instance Segmentation**. The team must source or curate a dataset of diverse food images to train a model that can successfully isolate individual food items on a cluttered plate. Following segmentation, the system must

perform **Fine-Grained Classification** to identify specific ingredients. Students are encouraged to explore **Prompting for Large Language Models (LLMs)** to assist in this process; by feeding visual descriptions or raw labels into a vision-capable LLM, the system can infer complex ingredients and hidden components that a standard classifier might miss.

The third phase involves **Quantitative Assessment and Health Mapping**. The team must develop a **Predictive Learning** pipeline to estimate portion sizes and caloric density from two-dimensional images. This involves using **Regression Analysis** to map the visual volume of identified food items to nutritional values. Crucially, the system must be grounded in authority; the team is required to integrate **World Health Organization (WHO) Guidelines** regarding dietary intake for specific health conditions, such as hypertension or diabetes. This ensures that the system's internal logic for "risk" is not arbitrary but is based on global medical standards for sugar, salt, and fat consumption.

Finally, the team must implement a **Multimodal Advice Engine**. This task involves using **Retrieval-Augmented Generation (RAG)** to fuse the visual data of the meal with the user's health goals and the relevant WHO evidence. The goal is to produce "Smart Advice" that is safe, supportive, and grounded in fact. By utilizing **LLM Prompt Engineering**, the team can ensure the chatbot translates complex nutritional data into clear, conversational suggestions. The final system should be housed in a functional **User Interface (UI)** that allows for seamless photo uploads and real-time dialogue, providing a complete end-to-end prototype of a digital nutrition coach.

4. Implementation Roadmap (13-Week Semester)

Phase	Weeks	Tasks
1	1–3	System Architecture, UI Design, and User Goal Definitions.
2	4–7	Computer Vision Model Training (Segmentation & Classification).
3	8–10	Regression Logic (Calories) and Chatbot Logic (NLP).
4	11–13	Integration, API Testing, and Final Prototype Validation, Demo, Report Writing.

5. Project Deliverables & Submission Requirements

To complete the project, the team must submit a professional-grade package that demonstrates technical depth and transparency. All components must be uploaded to the project portal before the final demonstration.

Deliverable	Requirements & Contents	Format
Technical Codebase	Full source code for the CV models, NLP chatbot, and integration scripts. Must include a README for setup.	GitHub/GitLab Repo
Project Report	A detailed paper covering architecture, ML/NLP branch selection, and model evaluation (accuracy, F1-scores).	PDF Document
Data & Evidence	Sourced food datasets, training labels, and the WHO-based knowledge used for the advice engine.	Data Folder / CSV
Presentation Deck	Slides covering the problem statement, technical hurdles, and the final value proposition.	PowerPoint / PDF
System Demo	A live walkthrough showing image upload, nutritional analysis, and user dialogue.	Live Prototype